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Thermally activated crack-propagation in brittle materials

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Abstract. Under completely brittle conditions a crack can only be stabilized by lattice trapping. The crack rate $\dot{a}$ *in vacuo* is then controlled by the rupture of individual bonds at the crack tip with the aid of thermal activation. A careful thermodynamic analysis shows that the thermodynamic activation area is somewhat larger but comparable in magnitude to the atomistic activation area (i.e., the area by which the crack advances in each activation event). By plotting the experimental results in $\dot{a}$ vs K^2 (K = stress intensity factor) it is possible to derive the activation area from the slope of the curve and hence to determine whether brittle conditions can prevail.

1. Introduction

When a solid containing a crack is stressed by an external loading mechanism there exists a critical stress at which the crack accelerates dynamically leading to catastrophic failure. Below this stress there is a range in which a slow viscous-like movement of the crack tip occurs. This has sometimes been called subcritical crack-growth. The reasons for this viscous-like behaviour are manyfold and are different in brittle and non-brittle materials. In non-brittle materials the viscous movement can occur because energy is dissipated by the emission of dislocations from the crack tip whereas in brittle failure the energy dissipation occurs by some thermally activated processes. Such brittle behaviour is observed for instance in Si, Ge, Al_2O_3, ceramics [1] and in glass where it has been studied in detail in the classical work of Wiederhorn [2]. When the rate of the crack tip is plotted against the stress intensity factor K (which is proportional to the stress times the square root of the crack length) there exist three regimes [2]. At very low K values the crack can only propagate when the energy needed to break atomic bonds at the crack tip is lowered by some chemical reaction with the gaseous or liquid medium surrounding the sample. At intermediate K values the crack velocity is practically constant and is determined by the diffusion rate of this chemically active medium towards the crack tip. At even higher K values or when the sample is in vacuo it is the thermal activation helping to break the bonds at the crack tip which control the crack velocity. In this regime the crack can be stabilized in a discrete atomic structure when "lattice trapping" occurs. This process has been proposed and studied in detail by Thomson and coworkers [3], though it has been noticed before in computer simulations by Gehlen and Kanninen [4].

We are going to discuss thermally activated crack propagation in the lattice trapping regime when the stress-assisted "rupture" of atomic bonds is the rate controlling process. In this case the activation energy is not a unique property of the material but depends strongly on the elastic driving force f_e acting on the crack. Generally $f_e = K^2/M$ where M is an elastic modulus. There exist other possibilities of crack advance by diffusive processes in the solid material ahead of the crack tip as realized for instance in creep rupture. In this case the driving force controls the diffusive flux but the activation energy is affected very little by the applied stress.

We will focus our attention on the range of parameters to be expected when stress assisted bond rupture occurs and how the crack-rate $\dot{a}$ is affected by the stress intensity K. In the literature there are numerous previous treatments of such thermally activated crack growth [5–10], however, the thermodynamic state functions were not always chosen correctly.

There exist some formal similarities with thermally activated dislocation movement [11] as already pointed out by Pollet and Burns [12]. There are, however, important differences which essentially result from the fact that a dislocation is a true lattice defect whereas a crack is a boundary problem. Therefore careful considerations are necessary to correlate the experimentally determined phenomenological activation parameters with the atomistic mechanism controlling the crack propagation rate. Under brittle conditions, when the crack rate is controlled by the thermally assisted breakage of individual bonds or the overcoming of local obstacles at the crack front, it is possible to give theoretical estimates of these activation parameters. By comparing the experimental results with the theoretical values it is possible to decide whether crack growth is controlled by lattice trapping.

2. Crack under lattice trapping

In order to define our thermodynamic system we assume an elastically isotropic infinite plate under a tensile stress σ clamped at infinity and in contact with a bath of constant temperature. With this boundary condition the equilibrium state is obviously determined by the minimum in free energy F which in the original (uncracked) state may be F_0. When a Griffith crack of length $2a$ is introduced through the plate perpendicular to the tensile axis the decrease in elastic energy F_e per unit thickness (which under isothermal conditions is a free energy) is given by the now well-known result

$$F_e - F_0 = -\frac{\pi a^2 \sigma^2}{M}. \tag{1}$$

(For a historical discussion see [13]). Here M is an elastic modulus. Under plane stress conditions $M = E$ where E is Young's modulus and under plane strain conditions $M = E/(1 + \nu^2)$ where ν is Poisson's ratio.

The choice of our system (Griffith crack, mode I, clamped sample) is convenient and does not lead to a serious loss of generality. Other crack configurations can be dealt with by introducing geometrical correction factors and other stress modes by introducing different elastic moduli. When the crack is introduced while the sample is dead loaded, the elastic energy in the sample would increase instead when the crack advances. In this case the appropriate thermodynamic potential would be the free enthalpy G which includes the potential energy of the external forces but the treatment would lead to identical results.

When the sample is completely brittle (i.e., when no dislocations are emitted from the crack tip during propagation) the only additional contribution to the free energy of the sample is the energy of the newly created surface F_s. Using the concept of a continuum with specific surface energy γ we then have per unit thickness

$$F_s = 4a\gamma. \tag{2}$$

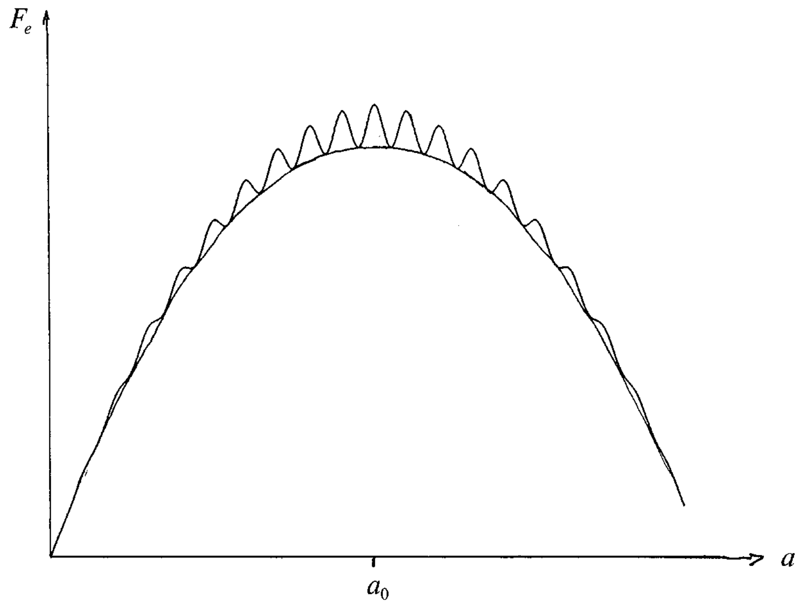

Fig. 1. Free energy of a clamped and stressed sample containing *a* crack as function of crack length *a*. A continuum gives smooth parabola, a discrete lattice a superimposed periodic variation. (Schematic).

The difference in free energy F per unit thickness of the system containing a crack and the free energy F_0 of the system without crack is

$$F - F_0 = F_s + F_e - F_0 = 4a\gamma - \frac{\pi a 2\sigma^2}{M}. \tag{3}$$

This dependence $F(a)$ in a continuum is shown in Fig. 1 as the smooth parabola. The corresponding driving force per unit thickness in our macroscopic description is given by

$$f = -\frac{\mathrm{d}F}{\mathrm{d}(2a)} = \frac{\pi a \sigma^2}{M} - 2\gamma, \tag{4}$$

where the first term on the right side is due to the release of elastic energy per unit length. This can be generalized to other configurations by introducing the stress intensity factor K in writing

$$f_e = -\frac{\mathrm{d}F_e}{\mathrm{d}(2a)} = \frac{K^2}{M}, \tag{5a}$$

$$K = \sigma(\pi a)^{1/2} Y, \tag{5b}$$

where Y is a correction factor characteristic for a specific geometrical configuration and (5a) is the general release rate of elastic energy. In our geometry $Y = 1$. For a side crack of length a at the surface of a semi infinite plate we would have for instance $Y = 1,12$ and for a penny shaped crack in an infinite medium under axial loading $Y = 2/\pi$. Values for Y in other geometrical configurations can be found in the literature [14].

The equilibrium configuration determined by $(\partial F/\partial 2a) = 0$ corresponds here to a maximum in F occurring at a critical half length $a_0 = 2\gamma M/\pi\sigma^2$ or to the Griffith condition $K_0^2 = 2\gamma M$. This leads to the well known result that in an elastic continuum a crack is unstable and will either shrink when $a < a_0$ or expand when $a > a_0$.

Stabilisation of the brittle crack can occur, however, when we take account of the discrete atomic structure of matter or when localized elastic inhomogeneities are present. Since a general quantitative treatment of the interaction of the crack with other localized defects is still missing we consider here in detail the effect of "lattice trapping" first discussed by Thomson et al. [3]. The thermodynamic formalism is, however, not limited to this case and can be generalized to any situation where the crack front interacts with localized obstacles. The origin of lattice trapping results from the fact that the total energy stored in the stressed atomic bonds in the immediate neighbourhood of the crack tip varies with the displacement x of the tip within one lattice cell. The position of the tip is here defined by the center of the elastic displacement and the resulting stress field far away from the crack front. This energy variation can also be demonstrated by computer modelling [4, 15, 16]. Figure 2 schematically shows a cut perpendicular to the front of a crack propagating along a close packed direction in a crystalline medium. Configuration I corresponds to a stable crack position x_1. After the stretched interatomic bond near the crack tip has been broken, the crack advances one atomic distance b to configuration III at x_3 which has identical atomic arrangement except for the fact that the free surface at the crack has increased by a pair of atoms. Between these two configurations there is a critical metastable configuration II at x_2 of higher energy called the saddle point configuration before bond rupture.

By the nonlinear action of the bonds the profile of the crack surface (which in a continuum would be parabolic) will be modified in a way discussed by Barenblatt [17] for a continuum. The essential aspect, however, is here that the energy density near the crack tip varies periodically due to the discrete atomic structure. These small energy variations are superimposed on the smooth parabola for the continuum, thus producing a dinosaurback curve [4] shown in Fig. 1.

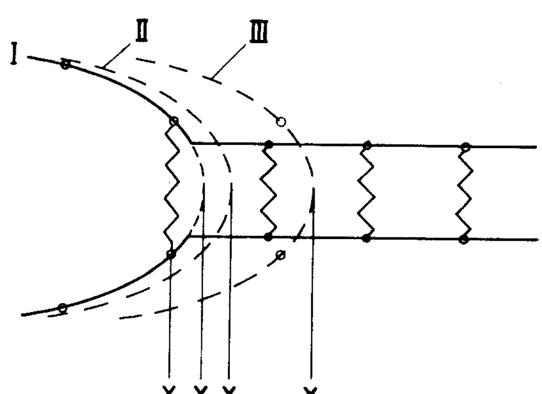

Fig. 2. Schematic atomic configuration near the crack tip in equilibrium I and III and saddle point configuration II.

3. Thermal activation

We are now interested in how the crack stabilized by lattice trapping can advance by overcoming these small energy variations with the aid of thermal activation. There are a number of different treatments of thermally activated processes [11] which apply for different conditions and differ in essential details. They all share, however, a common feature, which in its simplest form is the result at the transition state theory [18] which gives the rate Γ of a system overcoming an energy barrier as

$$\Gamma = \Gamma_0 \mathrm{e}^{-\Delta F/kT}, \tag{6}$$

where Γ_0 is the "attempt frequency" of the system oscillating in an equilibrium position in front of the barrier and ΔF the difference in free energy (for our system with fixed boundaries) between the equilibrium position in front of the barrier and the saddle point position "on top" of the barrier. The term exp $(-\Delta F/kT)$ is the ratio of the two corresponding partition functions and represents the thermo-dynamic probability that the system can be excited from a ground state to a state whose free energy is higher by ΔF.

Before discussing the evaluation of (6) we want to point out that a macroscopic system can only develop spontaneously in a direction in which the free energy decreases. This means that the crack can only grow when its length $2a$ exceeds the critical length $2a_0$. The term "subcritical crack-growth" is therefore somewhat misleading.

The problem of calculating Γ_0 and ΔF is a formidable task. The correct way to proceed [19] is to represent the free energy of the crystal in phase-space in which the coordinates are the momenta and the positions of all the atoms in the crystal and then to look for the path which leads from the equilibrium configuration I of Fig. 2 to configuration III over the smallest energy barrier corresponding to the saddle point configuration II.

In practice it suffices to calculate only the displacements of a limited number of atoms in the nonlinear region near the crack tip and to match the atomic positions on the surface of this region to the displacements resulting in an anisotropic elastic continuum from a crack with stress intensity K [20, 21]. When we restrict this atomistic region to the cleavage plane ahead of the top we have essentially the Peierls model of lattice trapping as used by Smith [22] and implicitly by Thomson et al. [3] and analytical solutions are possible. There exist, however, more realistic and very elaborate methods for computer modelling to determine numerically the positions of the atoms and the lattice energy in the region close to the crack tip [16]. The essential uncertainty results from our lack of knowledge of a correct interatomic potential. Since in these calculations the atoms relax to their static equilibrium positions the results correspond to a situation at temperature of $T = 0\,\mathrm{K}$.

We now let the crack advance while K stays fixed. Independent of the exact form of the interatomic potential, the atomic energy $U(x)$ in this region as function of the crack tip positon x must have the form shown in Fig. 3a whenever lattice trapping occurs. With each advance $\Delta x = b$ the atomic energy $U(x)$ increases due to the breaking of a bond exactly by the binding energy per unit length $U_B/b = 2\gamma b$. In relating U_B from interatomic forces to γ we actually have to allow for possible electron redistribution at the surface [23] and since γ is a free energy $U_B/2b^2$ is rather connected with $(\gamma - T\partial\gamma/\partial T)$. Superimposed on this continuous increase there is a periodically varying atomistic energy $g_K(x)$ contribution per

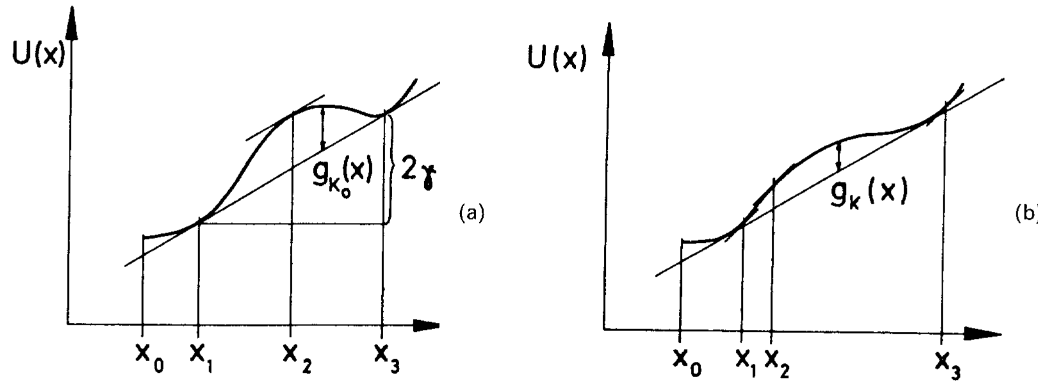

Fig. 3. Atomic energy $U(x)$ near the crack tip under constant K as the crack advances from equilibrium x_1 to saddle point x_2. (a) for $K = K_0 = (2\gamma M)^{1/2}$ (b) for $K > K_0$. The slope of $U(x)$ at x_1 and x_2 equals f_e.

unit length

$$U(x) = g_K(x) + 2\gamma x, \tag{7}$$

which is responsible for lattice trapping. Formally $g_K(x)$ can be considered as a periodically varying contribution to the surface tension [9, 10].

When the crack advances by an infinitesimal distance Δx the change ΔF_e in elastic energy of the system outside a boundary surface surrounding the crack tip is obviously equal to the flux of energy momentum through the surface as discussed by Eshelby [24]. This flux is independent of the position of the surface enclosing the inhomogeneity as long as in being shifted it does not cross another singularity. Since this surface has been chosen to lie in a region where linear elasticity holds this energy change is independent of nonlinear behaviour near the crack tip and obviously given from (5) as

$$\Delta F_e = f_e \Delta x = \frac{K^2}{M} \Delta x. \tag{8}$$

Equilibrium positions x_1 and the saddle point positions x_2 of the crack are only possible in a range where the elastic force f_e can be balanced by the "atomic force" or where the condition

$$\frac{\mathrm{d}U(x)}{\mathrm{d}x} = f_e \quad \text{for } x = x_1, x_2 \tag{9}$$

can be satisfied. When $K = K_0$ we have $f_e = 2\gamma$ and $x_1(K_0)$ and $x_2(K_0)$ are at the positions where the local slope of $U(x)$ is equal to the general slope (Fig. 3a). Increasing K will increase f_e and bring x_1 and x_2 closer to each other and trapping will disappear when f_e reaches the maximum value f_m corresponding to the maximum slope of $U(x)$ at the inflection point.

We now assume that in the crack front a segment of length l moves from the equilibrium position x_1 to the saddle point position x_2. The change in free energy can in principle be determined by integrating along the reaction path (under constant f_e)

$$\Delta F = l \int_{x_1(f_e)}^{x_2(f_e)} \left(\frac{\partial U(x)}{\partial x} - f_e \right) \mathrm{d}x = l[U(x_2) - U(x_1) - f_e(x_2 - x_1)], \tag{10}$$

where x_1 and x_2 both depend on the elastic force f_e.

A reliable value of ΔF can only be obtained when we have adequate information on the interatomic potentials which is rarely available. Even without such knowledge we can however determine the variation of ΔF with f_e using the same arguments as in the analysis of thermally activated dislocation movement [25]. The negative value of this derivative of the free energy of activation ΔF with respect to the force per unit length f_e is also called the thermodynamic activation area ΔA_{th} and we obtain from (10)

$$-\Delta A_{\text{th}} \equiv \frac{\mathrm{d}\Delta F}{\mathrm{d}f_e} = l \int_{x_1}^{x_2} \frac{\partial}{\partial f_e} \left(\frac{\partial U}{\partial x} - f_e \right) \mathrm{d}x + l \left[\frac{\partial U}{\partial x} - f_e \right]_{x_2} \frac{\mathrm{d}x_2}{\mathrm{d}f_e} - l \left[\frac{\partial U}{\delta x} - f_e \right]_{x_1} \frac{\mathrm{d}x_1}{\mathrm{d}f_e}, \tag{11}$$

where we have applied Leibnitz's rule for differentiation of integrals with variable limits. The values of the square brackets in the second and third term are taken at the starting position x_1 and at the saddle point x_2, respectively. Since, however, x_1 and x_2 are equilibrium positions these terms vanish and we are left with the first term in (11).

When now the atomistic energy $U(x)$ is a unique function of the position $x(f_e)$ of the crack tip and does not depend explicitly on f_e (or K) we have $\partial U/\partial f_e = 0$ and then it follows that

$$\frac{\mathrm{d}\Delta F}{\mathrm{d}f_e} = -l(x_2 - x_1) = -\Delta A, \tag{12a}$$

where ΔA is the atomistic area of crack advance between equilibrium and saddle point position. This would be analogous to plastic deformation when a dislocation overcomes a rigid obstacle, which is not modified by the applied stress.

Contrary to a dislocation whose stress field is determined by constant Burgers vector, the stress field of a crack is proportional to the stress intensity factor K (proportional to the applied stress) and hence variation in K will modify the atomic positions in equilibrium and hence modify $U(x)$ which is actually a functional $U_k(x)$ of K (as shown in Fig. 3b). A careful analysis of the resulting additional term in (11) is made in the Appendix and we obtain

$$-\frac{\mathrm{d}\Delta F}{\mathrm{d}f_e} = (\beta + 1)\Delta A, \tag{12b}$$

where ΔA again is the atomistic area of crack advance for $f_e = $ const. The value of β depends on the details of the trapping which in turn depends on the interatomic potentials. As shown in the Appendix for reasonable potentials and lattice stability we can estimate $0 < \beta \lesssim 2$ a value which is also obtained "experimentally" from computer modelling [16]. The reason why the thermodynamic activation area ΔA_{th}, is larger than the atomistic jump area ΔA lies in the fact that by increasing f_e not only the jump length but also the obstacle strength is decreased.

The calculation of the frequency factor Γ_0 is equally complicated [19]. It is actually the ratio of the product of all frequencies of the individual lattice modes in the saddle point divided by the same product taken in the equilibrium position. An additional complication arises since at the saddle point one of these modes becomes translational and converts from a lattice mode at x_1 to a surface mode at x_3. Considering the lack of our knowledge of the atomic parameters we can make a simplistic estimate based on localized modes of atomic vibration. When the crack front advances an interatomic distance b over the length l we would expect as a very rough estimate

$$\Gamma_0 \simeq \nu_D \frac{b}{l} \left(\frac{\nu_S}{\nu_D} \right)^n . \tag{13}$$

Here ν_D is the Debye frequency of the lattice modes and ν_S the corresponding frequency of the surface modes and $n = \approx l/b$ the number of atoms which are transferred from the bulk to the surface. With $\nu_S \approx \nu_D/2$ and assuming l not very much larger than b we obtain for Γ_0 a value about one order of magnitude smaller than but comparable to the Debye frequency ν_D.

4. The analysis of the crack rate

When a crack advances in the lattice trapping regime (i.e., by the consecutive breaking of individual bonds) we expect that after each thermal activation event a segment of length l moves over a distance $\Delta x < l$ both l and Δx of atomic dimensions. When Γ is the rate of formation of an individual jump and the crack has the length L the total rate of jump formation at the crack front is then $\Gamma L/l$. The area swept over by the crack per unit time is then $\dot{A} = (\Gamma L/l)\Delta x l$ and we have for the rate of crack advance $\dot{a} = \dot{A}/L$ the relation

$$\dot{a} = \Gamma \Delta x. \tag{14}$$

The available driving force f per unit length for each jump is determined by the gain in energy per jump and hence we have

$$f = f_e - 2\gamma \tag{15}$$

as discussed in (4).

When $f_e = 2\gamma$ or $K = K_0 = \sqrt{2M_\gamma}$ there exists equilibrium and for small f in principle back jumps would have to be considered. However, as discussed abundantly in the literature due to some irreversibilities they usually do not occur. Since the energy of the newly created surface $2\gamma b^2$ is typically of the order of 4×10^{-19} J and around room temperature $kT = 4 \times 10^{-21}$ J back jumps can usually be neglected when K exceeds K_0 by a few percent. Hence generally we have for the rate of crack advance

$$\dot{a} = \dot{a}_0 e^{-\Delta F/kT}, \tag{16}$$

with $\dot{a}_0 = \Gamma_0 \Delta x$. With the value of Γ_0 from (13) and Δx of atomic dimensions, we expect $\dot{a}_0$ somewhat less than the sound velocity i.e. between $10^3\,\mathrm{m\,s^{-1}}$ and $10^2\,\mathrm{m\,s^{-1}}$.

When observations are now made at a certain crack rate $\dot{a}$ the stress intensity K must be adjusted to a value which reduces the activation energy to

$$\Delta F(K) = -kT \ln \frac{\dot{a}}{\dot{a}_0}. \tag{17}$$

A typical experimental rate is $\dot{a} = 10^{-7}\,\mathrm{m\,s^{-1}}$ which will lead to $\Delta F(K) \approx 20\,kT$ in the lattice trapping regime.

On the other hand the atomistic area swept over during moving to the saddle point position should be of atomic dimensions and hence

$$\Delta A \approx cb^2, \tag{18}$$

with c ranging from 0.1 to 2 we would have $\Delta A \approx 10^{-20}\,\mathrm{m}^2$. When ΔA is considerably larger the obstacles for crack advance cannot be the breakage of each individual bond but rather the rupture of more widely spaced strong bonds.

Experimentally the crack rate $\dot{a}$ as function of K and T can be determined. Since $\Delta F = \Delta U - T\Delta S$ and $\Delta S = -(\partial \Delta F/\partial T)_v$ (for a clamped sample) we find using (17) and measuring the crack rate $\dot{a}$ at different temperatures the internal energy of activation

$$\Delta U = kT^2 \left(\frac{\partial \ln \dot{a}/\dot{a}_0}{\partial T} \right)_K. \tag{19}$$

A direct comparison between ΔU of (19) and ΔF of (17) is not possible without knowing the entropy change ΔS connected with the transition from the equilibrium position to the saddle point where bond breaking occurs. This entropy is connected with the change in the spectrum of the vibrational lattice modes due to nonlinear behaviour [26] when the crack moves from the equilibrium position to the saddle point and is extremely difficult to estimate. As first approximation we may use a phenomenological correlation between the entropy and the temperature dependence of the elastic modulus M [18] which yields $\Delta S = -(\Delta F/M)(\mathrm{d}M/\mathrm{d}T)$. Thus we obtain

$$\Delta F \approx \frac{\Delta U}{1 + \gamma}, \tag{20}$$

where $\gamma = -(T/M)(\mathrm{d}M/\mathrm{d}T)$. Since the relative decrease in the elastic modulus $\Delta MT/M\Delta T$ between 0 K and room temperature is typically 0.2 to 0.3 we expect at room temperature γ typically to have a value of this order of magnitude.

From experiment we can further determine an experimental activation area ΔA_e. When the pre-exponential factor $\dot{a}_0$ entering (17) does not depend on the stress intensity K, this ΔA_0 is identical with the thermodynamic activation area $\Delta A_{\mathrm{th}} = -\partial \Delta F/\partial f_e$. This condition should be satisfied in our case and then we have as discussed in (12b)

$$\Delta A_e \equiv -kTM \frac{\partial \ln \dot{a}/\dot{a}_0}{\partial K^2} = (1 + \beta)\Delta A. \tag{21}$$

When therefore ln $\dot{a}$ is plotted against $f_e = K^2/M$ the slope determines the experimental activation area ΔA_e which should be comparable to or up to a factor of 3 larger than the atomistic activation area ΔA. When therefore ΔA_e falls considerably below a value of 10^{-20} m^2 as estimated in (18) very probably lattice trapping can be ruled out as process controlling the crack rate.

5. Crack movement by the kink mechanism

In our treatment of crack propagation we have assumed that segments of the crack front with a limited length l of atomic dimensions advances by individual jumps. In an original straight crack front bulges would be formed by those jumps of the height of one interatomic distance and the question is how this formation of a bow-out affects the crack rate.

There exists some formal similarity with the formation of kink pairs in straight dislocations running along a close packed direction and crossing from one Peierls valley to the next one. Such a kink model for the propagation of the crack front has been discussed in a number of papers [4, 5, 9, 16, 27–29] and it has been argued in analogy to the movement of dislocations that the resistance to crack propagation should be lowered considerably by this mechanism. Again, however, this analogy is misleading. A kink is a stable eigenstate of a dislocation (actually an excited soliton state [30]) which has a higher self-energy. Once a kink has been formed it can move readily sideways under the action of an applied stress and will undergo conservative movement (except for very small periodic energy variations with the periodicity of the lattice).

Whether the formation of a flat bulge in a crack front releases an energy different from the one which is obtained by advancing the crack front parallel to itself is still not decided but arguments can be put forward [24, 31] that the energy is the same as long as the height of the bulge is small compared to the local curvature at the crack front. This energy is given by $\Delta A \cdot K^2/M$ where K and M have local values, and ΔA the area swept over.

Once a bulge has been formed in the crack front by breaking an atomic bond the side-way movement of this bulge occurs nonconservatively by forming adjacent bulges which require the same activation and which will release the same energy (and hence experience the same driving force) as the original bulge. The only difference is that when an original bulge is formed regions of increased and decreased curvature (with respect to the direction of crack advance) will occur. As Rice [32] has shown recently for a crack front with a sinusoidal disturbance the K value of the leading part is decreased and of the trailing part increased when the wave length is small. This effect will have a tendency to stabilize a straight crack front and will help to spread the bulge sideways. When the height of the bulge is small against its length the effect should be small.

Superimposed on this variation in K due to curvature there exists an anisotropy in the energy release rate because M shows an orientation dependence for mode II and mode III cracks and in anisotropic media even in mode I [20, 33], but again the effect should be small and could have either sign.

There is, however, another effect which should favour crack propagation in the trapping regime in crystalline solids when the dislocation is at some angle with respect to a close packed direction. The same effect occurs when a reasonably stiff dislocation moves through a square array of very localized point obstacles. The distance between the obstacles contacted by the dislocation is larger in a high index direction and hence the force on each obstacle will also be larger. The breakage of bonds is such a highly localized event and hence the width l of the bulge which enters in the activation area should increase in some low index direction. Therefore the area ΔA in moving up to the saddle point should increase, whereas the energy to break the bond remains the same. But again due to the flexibility of the crack front the effect should be small and therefore contrary to dislocations we do not expect that "kink formation" (or "jog-formation") in crack fronts will have dramatic effects on crack propagation.

6. Discussion

There are a large number of more or less similar theories of thermal activation of crack growth in brittle materials, many of them have been discussed for instance by Wiederhorn [34] and Brown [6]. They can be divided essentially in two groups which differ by the way they treat the stress dependence of the free activation enthalpy ΔG which is assumed to be either a linear or a quadratic function of the stress intensity factor K.

We should again point out that since we have treated a clamped sample our driving force results from a release of stored elastic energy E_e and under isothermal conditions we have $\Delta F_e = -\Delta E_e$. In a dead loaded sample the elastic energy increases by ΔE_c upon crack advance. In this case the behaviour of the system is determined by the change in free enthalpy $\Delta G = \Delta E_e + \Delta W_a$ where ΔW_a is the change in potential of the external loading mechanism. Since when the crack advances infinitesimally $\Delta W_a = -2\Delta E_e$, the treatment of the dead loaded sample leads to the same conclusion. This result is actually a consequence of Betti's reciprocal theorem.

The theories which use a linear dependence of free enthalpy ΔG on K are formulated in formal analogy to the thermodynamic of gases under constant pressure with enthalpy $\Delta H = \Delta U + p\Delta V$. In the term $p\Delta V$ which represents the work done at the system by the pressure p of the gas is then replaced by the stress σ acting on the solid. This corresponds to assuming that when the crack advances a constant stress intensity K acts on the critical bond at the crack tip and the activation volume ΔV is then identified with the stretching at this bond leading sometimes to unreasonably large values [8].

The fallacy of this argument lies in the fact that in reality when the crack advances both the stress intensity K and the displacement at the position of the critical bond change simultaneously and this leads to a driving force proportional to K^2. This driving force was derived in (5) by considering the change in energy density in the whole system but Irwin [35] has shown that it also can be obtained locally as the work done during the displacement of the tip of a Griffith crack. Barenblatt [17] has extended these considerations to situations when the shape of the crack tip is modified by some interatomic forces and both treatments agree when these interatomic forces show a nonlinear behaviour [36].

There are a number of theories [9, 10, 13, 24] which recognize the fact that it is the energy release rate f_e from (5) which enters into the activation enthalpy. They all assume then that in the region close to $K = K_0$ the enthalpy can be represented by linear function of f_e by writing $\Delta H = \Delta H_0[1 - K^2/K_0^2]$ which in analogy to the consideration leading to (17) would give a linear relation between $\ln \dot{a}$ and K^2 with constant slope $\Delta H_0/K_0^2$.

Most experimental data are usually plotted as $\ln \dot{a}$ vs K leading often to nearly linear relations. As we have shown above it is more meaningful to plot $\ln \dot{a}$ vs K^2 since the slope of this curve is proportional to the activation area ΔA. As we have seen this ΔA will generally decrease with increasing K. Therefore the desire of many experimentalists to find a linear relation has no physical foundation and deviations from linearity gives information on the bond strength. Actually many experimental data were made over such a narrow range of $\dot{a}$ and K values that within the considerable experimental scattering they can be represented by straight lines. Since $\partial \ln \dot{a}/\partial K^2 = (1/2K) \ln \dot{a}/\partial K$ a plot of $\ln \dot{a}$ against K will however, have the tendency to straighten out the decreasing slope of $\partial \ln \dot{a}/\partial K^2$.

The discussion given above shows the range of experimental parameters which we would expect when lattice trapping occurs. When for instance the experimental data of Wiederhorn et al. [37] on fracture of glasses in vacuum are analysed in this way we obtain in most cases ΔA values of the order of $10^{-20}\,\mathrm{m}^2$ and $\Delta F \approx 20\,\mathrm{kT}$ in the range of the theoretical values which we would expect from (17) and (18). This confirms the suggestion of these authors that crack growth in these materials is controlled by lattice trapping in contrast to other theories which assumed that alkali–ion diffusion or viscous flow ahead of the crack tip determines the crack rate. The original analysis of the experimental results [37] was made, however, by assuming a linear dependence of ΔF on K which as we have shown does not describe the behaviour in the lattice trapping regime correctly.

We finally want to point out that the same considerations can be applied in an analogous way to any crack movement which takes place by thermal activation over more widely spaced localized obstacles. Such localized obstacles could be for instance some strong bond configurations due to solute atoms or due to alloying compounds in glass or in ceramics. Such mechanisms can be identified when the activation area ΔA takes up values which exceed considerably the area corresponding to one atomic bond.

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Appendix I

In order to assess the contributions to the thermodynamic activation area $\Delta A_{\mathrm{th}} = -\mathrm{d}\Delta F/\mathrm{d}f_e$ we have to distinguish two potential curves (Fig. A1). During thermal activation the crack advances under the condition $K = \mathrm{const}$. Let us assume $K = K_i$ or $f_e = f_i = K_i^2/M$. When the crack moves in phase space from the equilibrium position x_1 to the saddle point x_2 it passes through non-equilibrium positions with a higher potential energy $U_i(x)$ and we can define a fictitious force $\varphi_i(x) = \partial U_i/\partial x$ which acts at the crack tip. Obviously we have $\varphi_i(x_1) = \varphi_i(x_2) = f_i$ (Fig. A1). Since the state of the bonds ahead of the crack tip is influenced by K we have generally for every K_i a different potential energy $U_i(x)$. On the other hand when we increase f_e the equilibrium position x_1 of the crack will generally also advance and we have now an atomic energy $U(x_1)$ where equilibrium demands $\mathrm{d}U(x)/\mathrm{d}x = f_e(x_1)$).

For not too exotic interatomic potentials we can assume that application of a higher elastic force f_e will lower the height g_K of the periodic variations in the atomic potential (Figs. 3a, 3b). This situation is assumed in Fig. A1 where we have represented $\varphi(x)$ and $f_e(x_1)$ in the

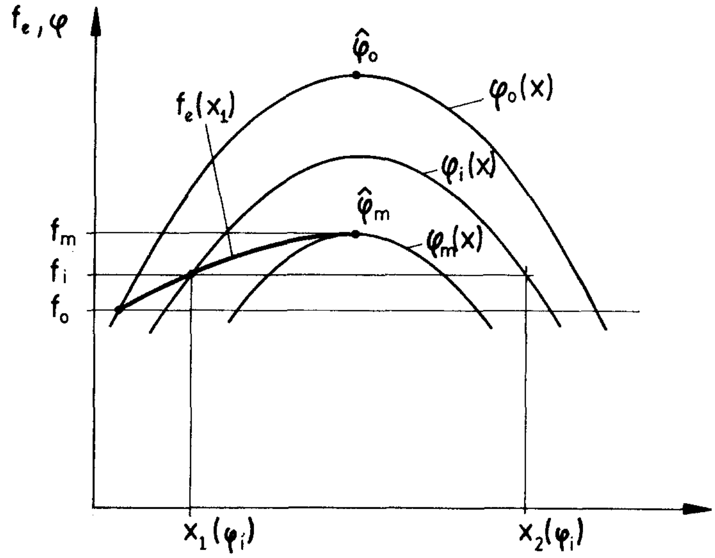

Fig. A1. Schematic representation of $\varphi(x) = (\partial U/\partial x)_{f_e}$ with f_e = const. and the relation $f_e(x_1)$ between the elastic force f_e and the equilibrium position x_1.

trapping range between $f_0 = 2\gamma b$ and an upper f_m. We define now the following relations for the maximum values of f and φ

$$f_0 = 2\gamma b \quad \hat{\varphi}_0 = pf_m,$$

$$f_m, = nf_0 \quad \hat{\varphi}_m = f_m,$$

where we have $n > 1$ and $p > 1$. The constant n defines the range of lattice trapping f_0/f_m which in principle can be found from experiments. Computer modelling has shown n to vary between 1.2 and 2 for realistic potentials in silicon [14] but it can have larger values when a linear spring type potential with bond snapping is used [27]. The constant p characterises the maximum fictitious force of lattice trapping to advance the crack one atomic distance when the external driving force is $f_0 = 2\gamma_0$. It is difficult to estimate. When the trapping range is small i.e. f_m close to f_0 the bond modification close to the tip should not change too much. Then we expect φ_0 close to φ_m and hence p close to 1. On the other hand even for large trapping p should hardly exceed 2. In the region around the crack position x where the largest forces are felt by the crack we then have

$$\frac{\Delta\varphi}{\Delta f_e} = \frac{\varphi_0 - \varphi_m}{f_0 - f_m} = -\frac{(p-1)n}{n-1} = -\beta.$$

This coefficient $\Delta\varphi/\Delta f_e$ represents an average numerical value for the coefficient $\partial^2 U(x)/\partial x\, \partial f_e$ occurring in the first term in (11). Then we obtain by integration of (11)

$$\Delta A_{\text{th}} = -\frac{\mathrm{d}\Delta F}{\mathrm{d}f_e} = (\beta + 1)\Delta A,$$

where $\Delta A = l(x_2 - x_1)$ is the atomic area swept over during the activation event. With the numerical estimates given above we expect $0 \leqq \beta \leqq 2$.

The validity of our arguments is actually borne out by the computer modelling of Sinclair in silicon [16] who determined $\Delta F(K)$ and independently the equilibrium and saddle point positions $x_1(K)$ and $x_2(K)$ as function of the stress intensity K. Comparison shows that β ranges between 1.5 and 2 for the various potentials used in the calculations. Under realistic conditions the factor $(1 + \beta)$ should vary therefore between 1 and 3.

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Résumé. Sous des conditions totalement fragiles, une fissure ne peut être stabilisée que par un piégeage du réseau. La vitesse de la fissure dans le vide est alors controlée par la rupture des liaisons individuelles à la pointe de la fissure, avec le concours de l'activation thermique. Par une analyse thermodynamique soigneuse, on montre que l'aire d'activation thermique est quelque peu plus grande, encore que comparable en taille, à l'aire d'activation atomistique (superficie dont progresse la fissure lors de chaque évènement d'activation). En portant les résultats expérimentaux de vitesse de fissure en fonction de K^2 (K = facteur d'intensité des contraintes), il est possible de déduire l'aire d'activation à partir de la pente de la courbe et, dès lors, de déterminer quelles conditions fragiles peuvent prévaloire.